Customer & Product Sales Analysis Report (2010–2013)

Prepared By

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Data Analyst Portfolio Project
Tools Used: SQL, Power BI, Excel

Contents

1. Executive Summary

This project focuses on analyzing customer purchasing behaviour, product performance, and overall sales trends between 2010 and 2013. The analysis was conducted using SQL for data extraction and transformation and Power BI for visualization and dashboard creation.

The primary objective of this project was to identify revenue-driving products, high-value customers, sales trends, and regional performance to support business decision-making.

The analysis revealed that a small percentage of customers contributed significantly to overall revenue, certain product categories consistently outperformed others, and sales experienced strong year-over-year growth during the analyzed period.

The final dashboard provides insights into customer behaviour, sales performance, product contribution, and regional profitability.

2. Business Problem Statement

Businesses generate large volumes of transactional data daily, but without proper analysis, it becomes difficult to identify customer trends, profitable products, and business growth opportunities.

The company wanted to:

- Identify top-performing products and categories
- Understand customer purchasing behaviour
- Track sales growth over time
- Analyze regional sales performance
- Improve business decision-making using data-driven insights

3. Project Objectives

The main objectives of this analysis were:

1. Analyze sales performance from 2010–2013
2. Identify top customers contributing to revenue
3. Discover best-selling and low-performing products
4. Understand sales trends and seasonal patterns
5. Evaluate regional performance and profitability
6. Build an interactive dashboard for business reporting

4. Dataset Information

Attribute	Details
Dataset Type	Customer & Product Sales Data
Time Period	2010–2013
Data Source	Public Business Dataset
Tools Used	SQL, Power BI, Excel
Main Tables	Orders, Customers, Products
Data Type	Transactional Sales Data

The dataset includes:

- Customer information
- Product details
- Sales transactions
- Product categories
- Order values
- Customer segmentation
- Purchase dates

The analysis was performed for:

- 2010
- 2011
- 2012
- 2013

5. Data Preparation Process

Data cleaning and preparation were performed before analysis to ensure data accuracy and consistency.

Steps Performed

1. Removed Duplicates

Duplicate transactional records were identified and removed.

2. Handled Missing Values

Null and blank values were checked and cleaned where necessary.

3. Standardized Data Types

Date columns and numerical columns were converted into appropriate formats.

4. Feature Engineering

Additional columns were created for analysis purposes:

- Revenue Categories
- Customer Segments

5. Data Validation

Cross-checks were performed to verify totals and ensure consistency across tables.

6. Database Exploration

The database structure was analyzed to understand table relationships and key business entities.

Tables Used:

Table Name	Description
Orders	Contains transactional sales information
Customers	Customer-related information
Products	Product and category information

Relationship Analysis

- Customer ID was used to connect the customer and order tables.
- Product ID was used to connect the product and order tables.
- Order Date was used for trend analysis.

7. SQL Analysis

SQL was used extensively for querying, cleaning, transformation, and analysis.

SQL Concepts Used

- SELECT Statements
- WHERE Conditions
- GROUP BY & HAVING
- JOINS
- CASE Statements
- Common Table Expressions (CTEs)
- Window Functions
- Aggregate Functions
- Ranking Functions
- Views

8. Exploratory Data Analysis (EDA)

EDA was conducted to identify patterns, trends, and business insights.

8.1 Data Exploration

~ Understanding the overall dataset structure.

Includes:

- Number of rows and columns
- Table names
- Null values
- Duplicate records

8.2 Dimensions Exploration

~ Analyzing categorical fields used for grouping and segmentation.

8.3 Date Exploration

~ Understanding the scope of data & Timespan

8.3 Measure exploration

~ Analyzing numerical business metrics.
Includes:

- Total Sales
- Profit
- Quantity
- Average Order Value

8.4 Magnitude

~ Compare the measure values by categories. It helps us understand the importance of different categories

8.5 Ranking

~ Order the value of dimensions by measure. Top N performance/bottom N performance

9 Advanced Data Analysis

9.1 Change over time trends

- Analyse how a measure evolves.

9.2 Cumulative Analysis

- Aggregate the data progressively.

9.3 Performance Analysis

- Comparing the current value to a targeted value

9.4 Part to Whole Analysis

- Analyse how an individual part is performing compared to the overall.

9.5 Data segmentation

- Grouping the data based on a specific range

9.6 Reporting

- Reporting dashboards were developed in Power BI to present business KPIs, sales trends, customer insights, and product performance in an interactive and visually understandable format.

10. Dashboard Overview

An interactive Power BI dashboard was developed to visualize business insights effectively.

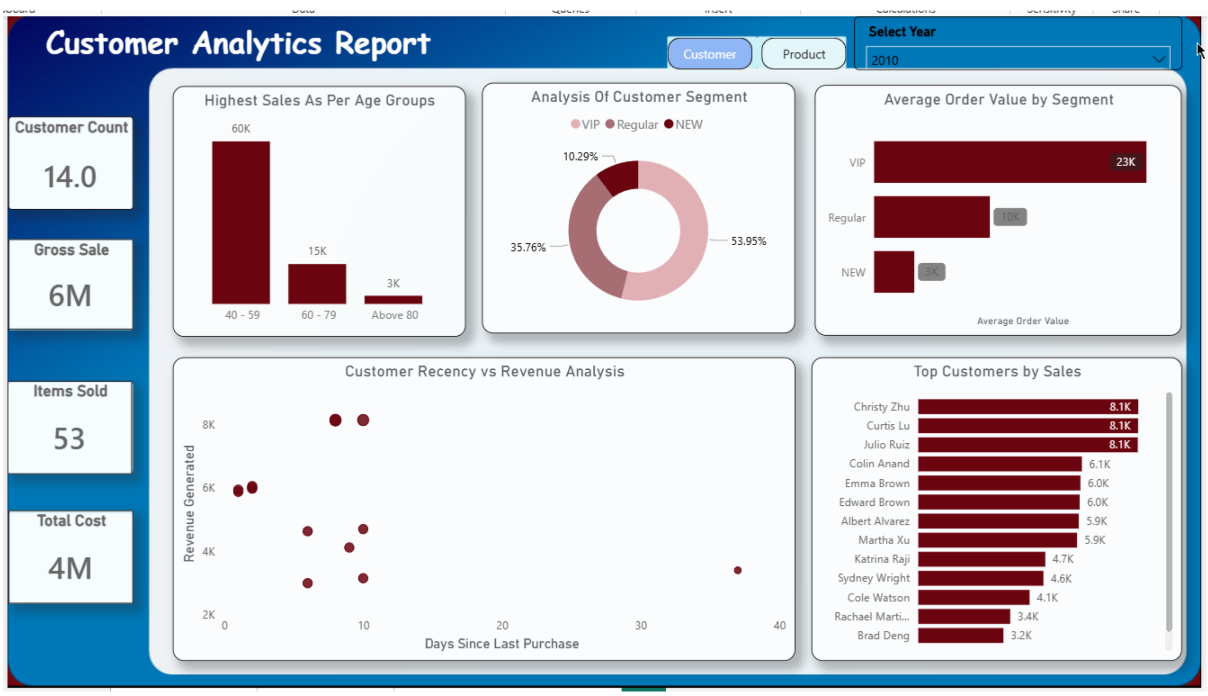
Key KPIs Included:

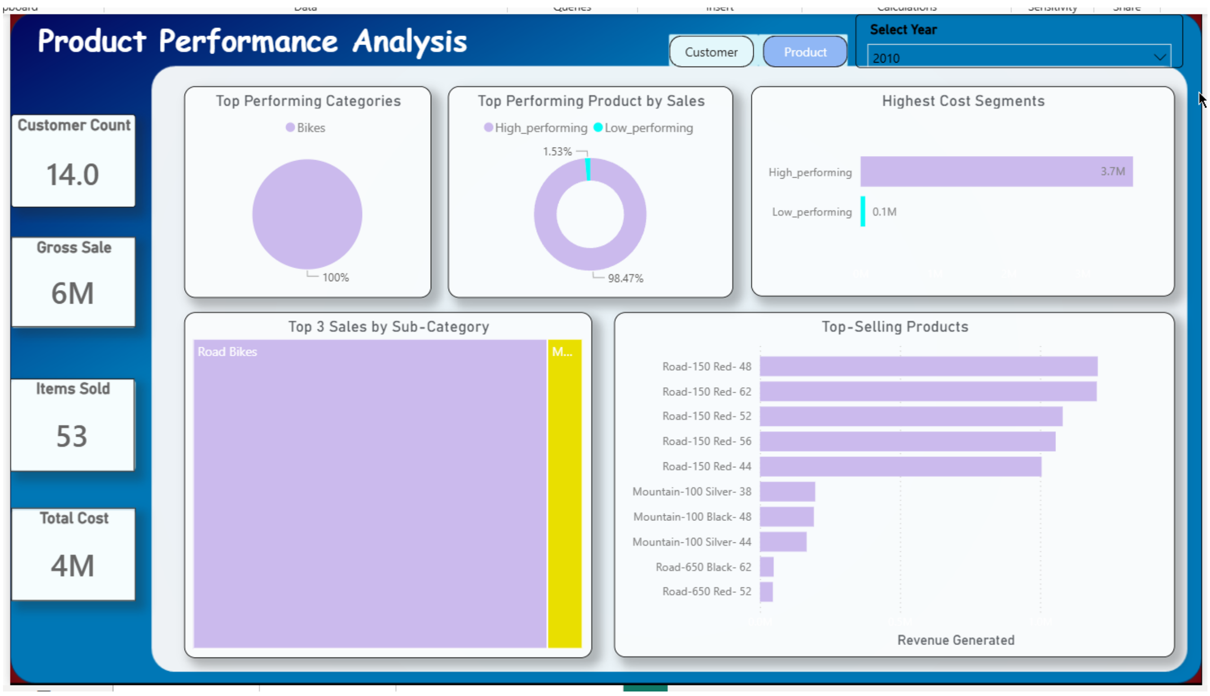
KPI	Description
Total Sales	Overall revenue generated
Total Profit	Net business profit
Total Customers	Number of unique customers
Total Orders	Number of transactions
Average Order Value	Average revenue per order

10.1 2010 Analysis — Premium Customer Foundation

❖ Business Performance

- The business generated **6M in gross sales** with only **14 customers**, indicating a very high-value customer base during the early stage.
- Profitability appears strong because sales significantly exceeded total costs (**6M sales vs 4M cost**).
- Product demand was still concentrated and niche-focused, with only **53 items sold**, suggesting a premium-product strategy rather than mass-volume sales.





❖ Customer Behaviour Analysis

- Customers aged **40–59** were the dominant revenue contributors. This indicates the business initially attracted mature and financially stable buyers.
- The **VIP segment accounted for 53.95%** of the customer base, showing that the company relied heavily on loyal high-value customers.
- Low recency (<10 days) suggests customers were purchasing frequently, indicating strong customer satisfaction and early loyalty development.

Strategic Insight:

This year reflects a **premium customer-driven business model**, where a small number of loyal customers generated most of the revenue.

❖ Product Performance Analysis

- Bikes were already the strongest-performing category and established themselves as the core business driver.
- **Road Bikes dominated sales**, especially:
  - Road-150 Red-48
  - Road-150 Red-62
  - Road-150 Red-52
- High-performing products also carried high operational costs, showing the business was investing heavily in premium inventory.

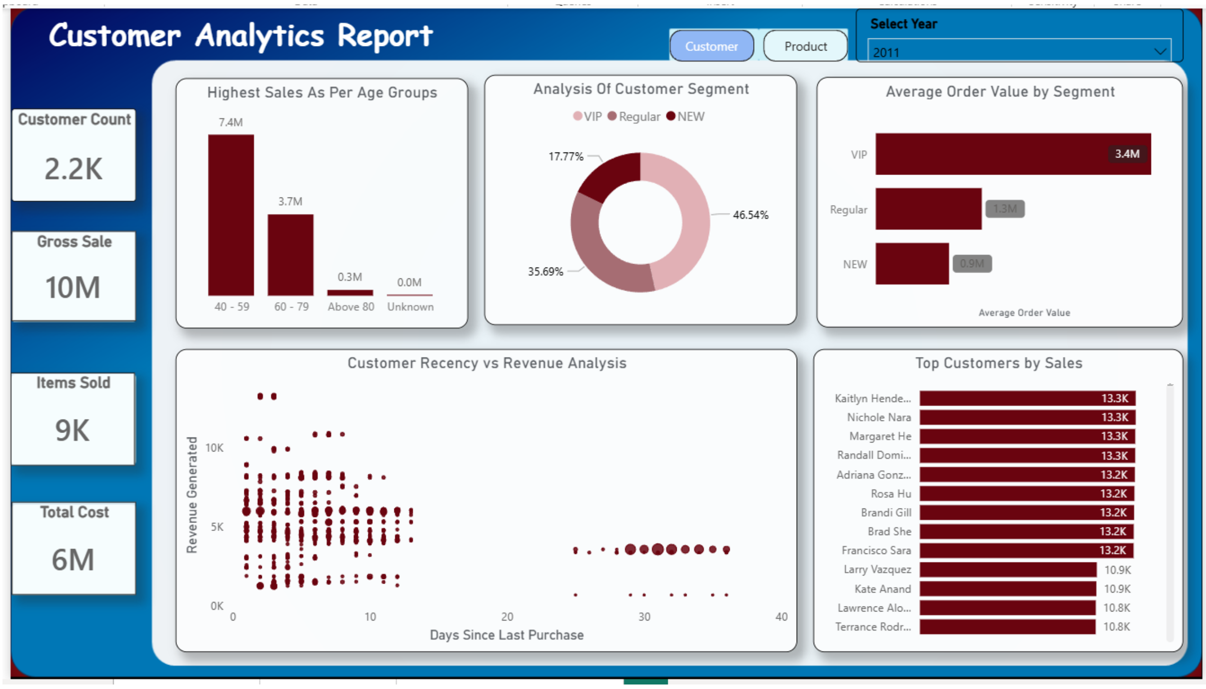
Strategic Insight

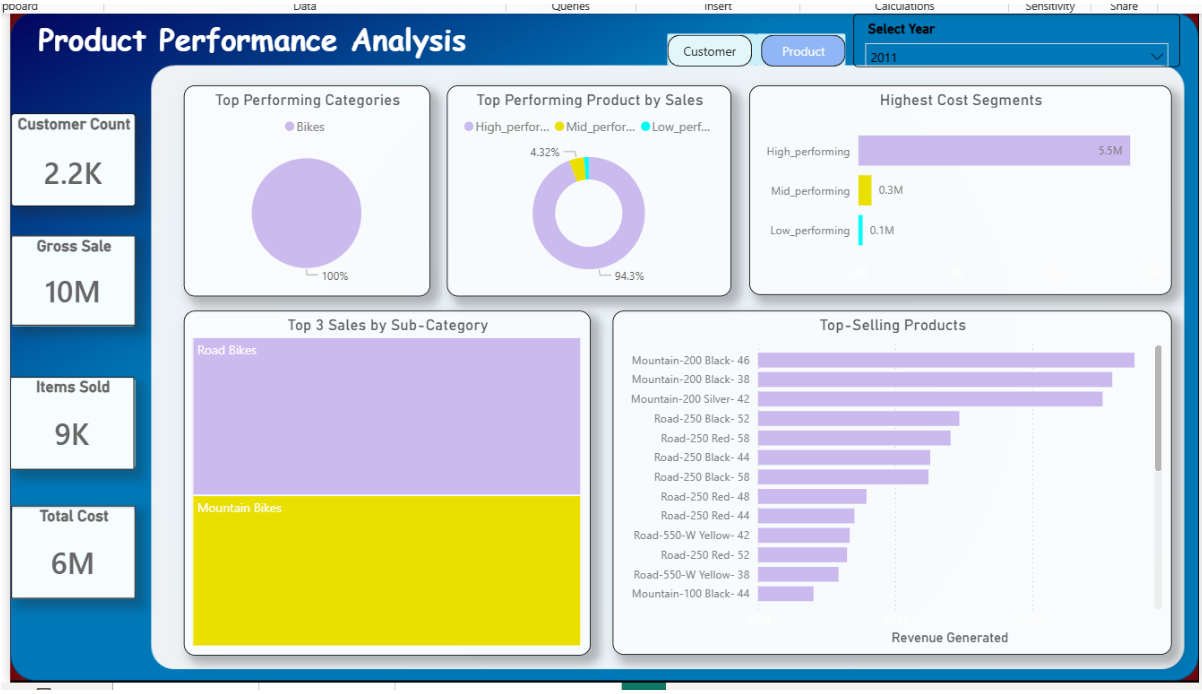
The company was strongly positioned in the **high-end road bike market** during 2010.

10.2 2011 Analysis — Growth & Expansion Year

❖ Business Performance

- Customer count increased dramatically from **14 to 2.2K**, showing rapid business expansion.
- Gross sales increased to **10M**, reflecting successful market penetration.
- Items sold increased sharply to **9K**, indicating a transition from premium-only selling toward higher transaction volume.





❖ Customer Behaviour Analysis

- The 40–59 age group continued to dominate revenue contribution, confirming a stable target demographic.
- VIP customers still remained the largest segment at **46.54%**, but the increase in new customers (**17.77%**) showed customer acquisition efforts were working.
- Customer recency remained healthy (<13 days), showing strong retention despite rapid growth.

Strategic Insight:

2011 represents the company's **scaling phase**, balancing customer acquisition while maintaining loyalty from existing premium customers.

❖ Product Performance Analysis

- Bikes remained the top-performing category.
- A major shift emerged:
  - **Mountain Bikes began competing strongly against Road Bikes**.
- Best-selling products shifted toward Mountain Bike models:
  - Mountain-200 Blank-46
  - Mountain-200 Blank-38
  - Mountain-200 Silver-42

Strategic Insight:

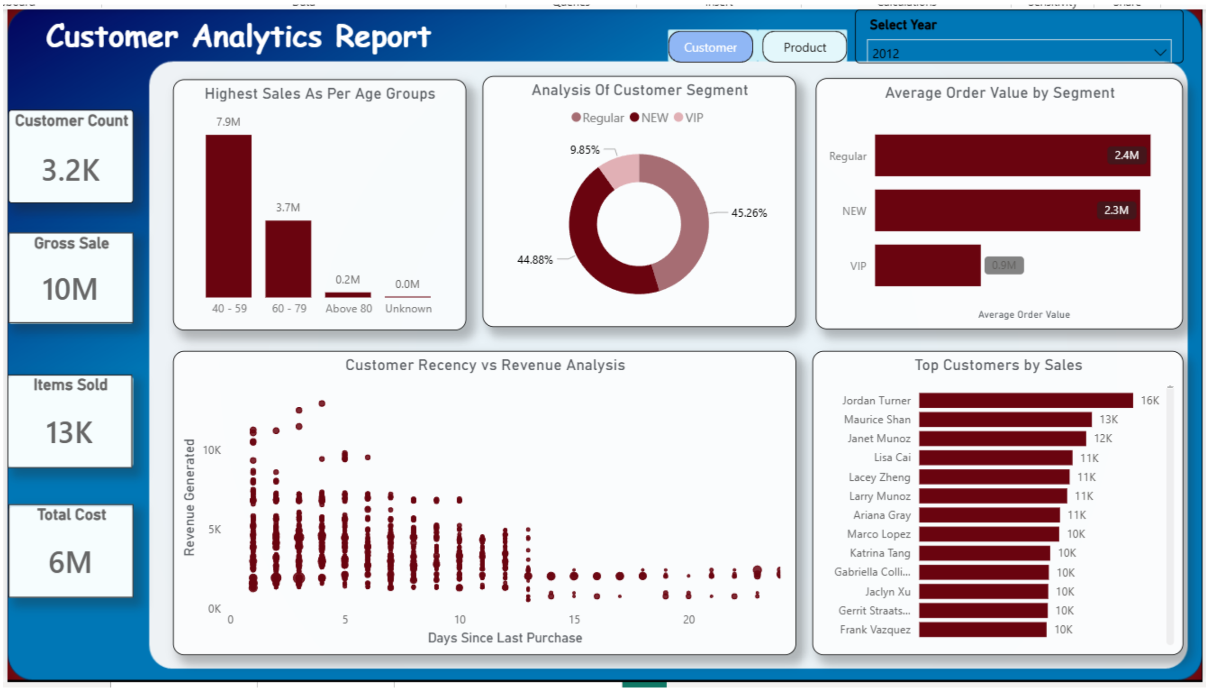
The market started showing changing customer preferences from performance road bikes toward more versatile mountain bikes.

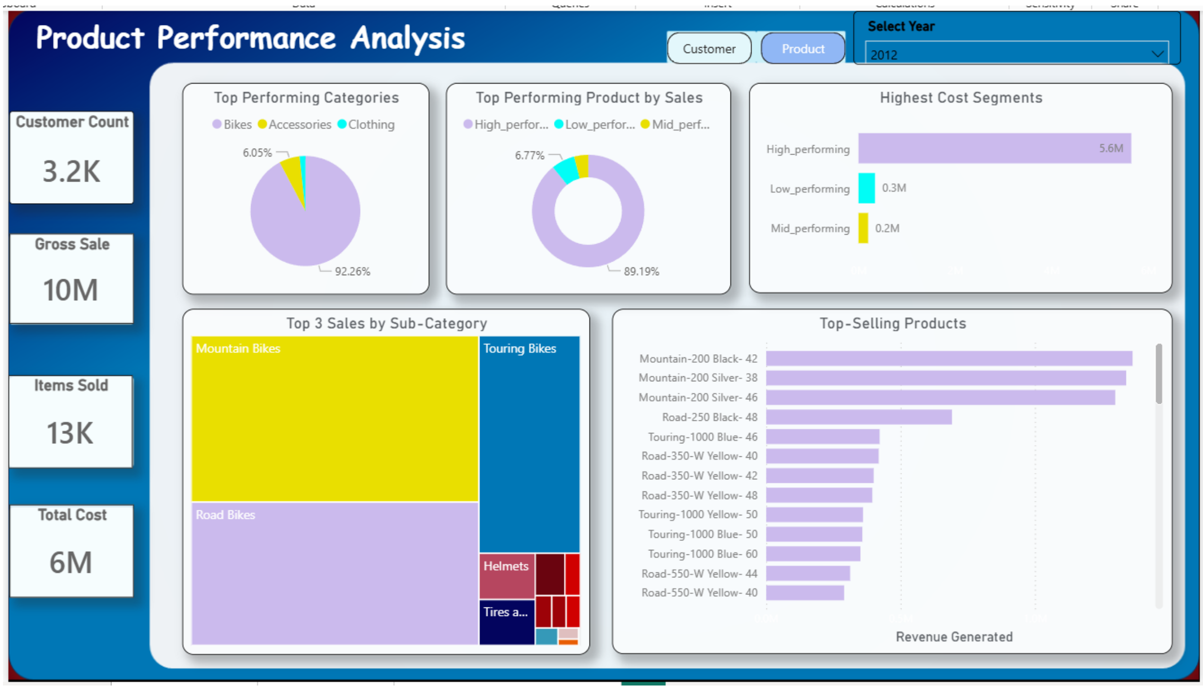
10.3 2012 Analysis — Market Transition Year

❖ Business Performance

- Customer count increased further to **3.2K**, showing continued growth.
- Sales remained stable at **10M**, despite a large increase in items sold.
- This suggests:
    - Average product prices may have reduced

The business may have shifted toward higher-volume selling





❖ Customer Behaviour Analysis

- A major transformation occurred in customer segmentation:
  - **Regular customers became the dominant segment (45.26%)**
  - New customers were almost equal (44.88%)
  - VIP contribution dropped sharply to 9.85%

Strategic Interpretation

This indicates the business moved away from relying only on elite customers and became more mainstream.

- Regular customers now generated the highest average order value.
- Customer retention still remained strong due to low recency (<13 days).

Strategic Insight:

2012 reflects a **customer base diversification phase**, where revenue became more evenly distributed across broader customer groups.

❖ Product Performance Analysis

- The biggest product transition happened in this year:
  - **Mountain Bikes overtook Road Bikes** completely.
- Top-selling products were now fully mountain-bike focused:
  - Mountain-200 Blank-42
  - Mountain-200 Silver-38

- Mountain-200 Silver-46

Strategic Insight:

The company successfully adapted to changing market demand by shifting inventory and marketing toward mountain bike products.

10.4 2013 Analysis — Mass Customer Acquisition Year

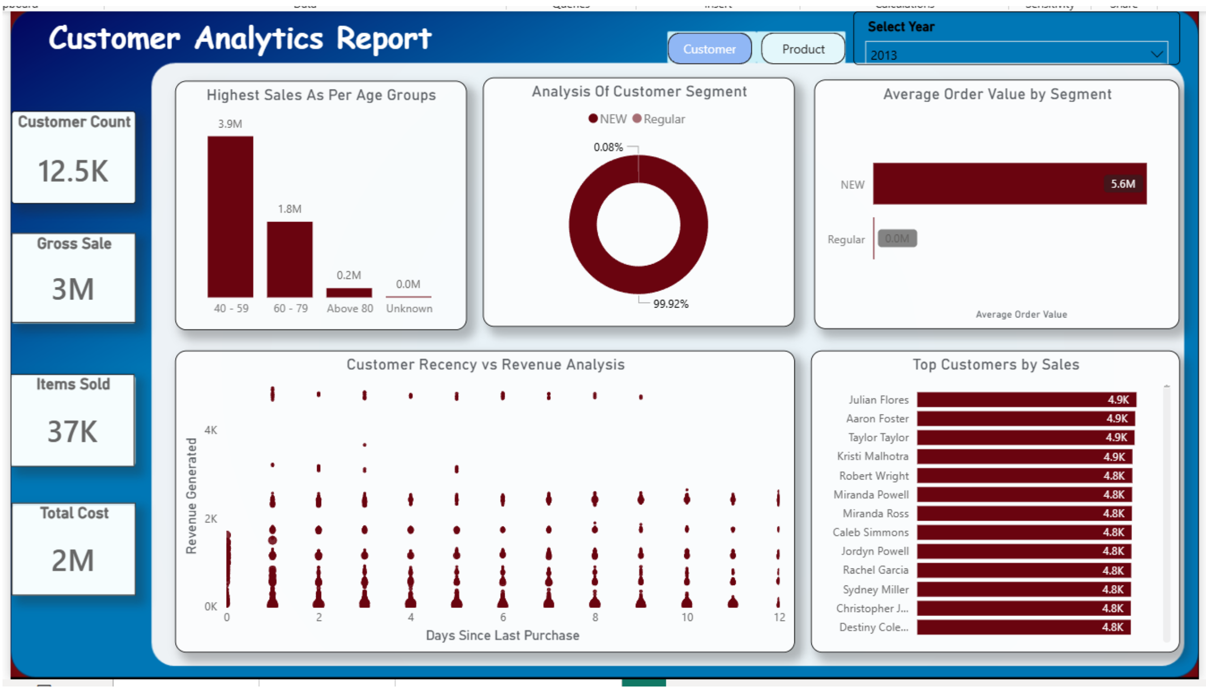
❖ Business Performance

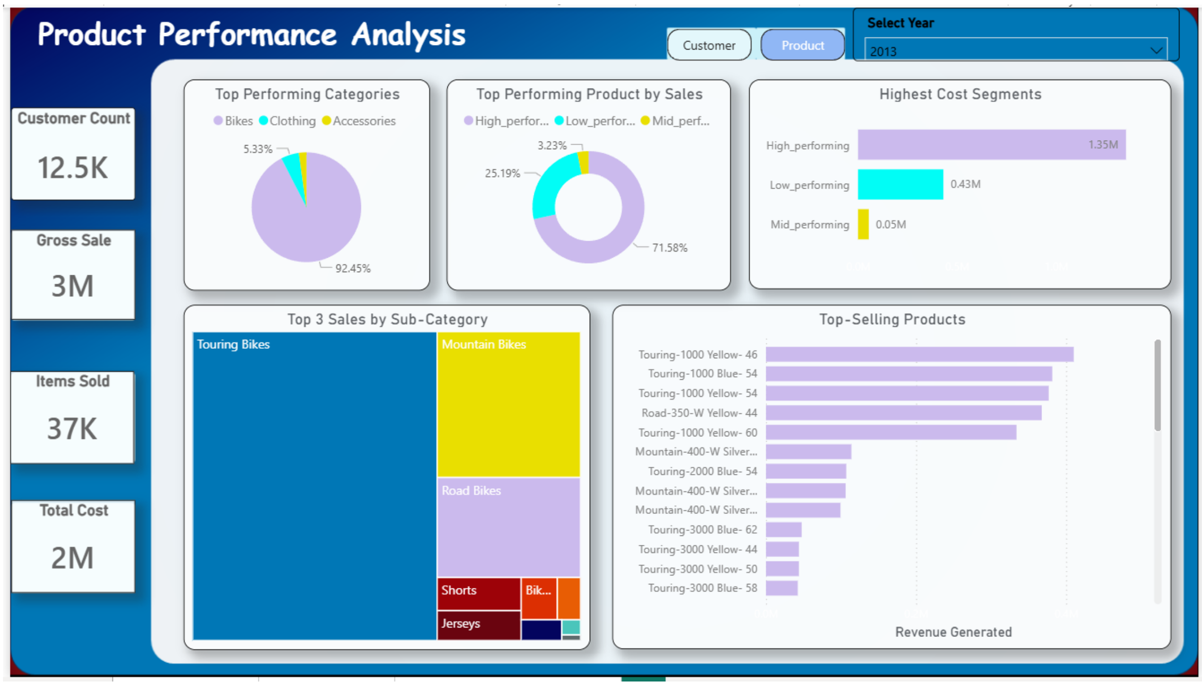
- Customer count exploded to **12.5K**, representing the highest growth across all years.
- However, gross sales dropped significantly to **3M**, despite:
    - 37K items sold

Strategic Interpretation

This indicates:

- The business likely shifted toward lower-priced products
- Profit margins may have reduced
- Revenue per customer decreased significantly





❖ Customer Behaviour Analysis

- The customer base became almost entirely composed of **new customers (99.92%)**.
- Regular customers dropped to almost zero.

Strategic Interpretation

This may indicate:

- Aggressive expansion campaigns
- Weak retention strategy
- Loss of loyalty customers
- Market repositioning toward acquisition instead of retention

Even though recency remained healthy (<12 days), the business may have struggled to convert new users into loyal long-term customers.

❖ Product Performance Analysis

- Touring Bikes became the dominant product segment.
- Top products:
    - Touring-1000 Yellow-46
    - Touring-1000 Blue-54
    - Touring-1000 Yellow-54

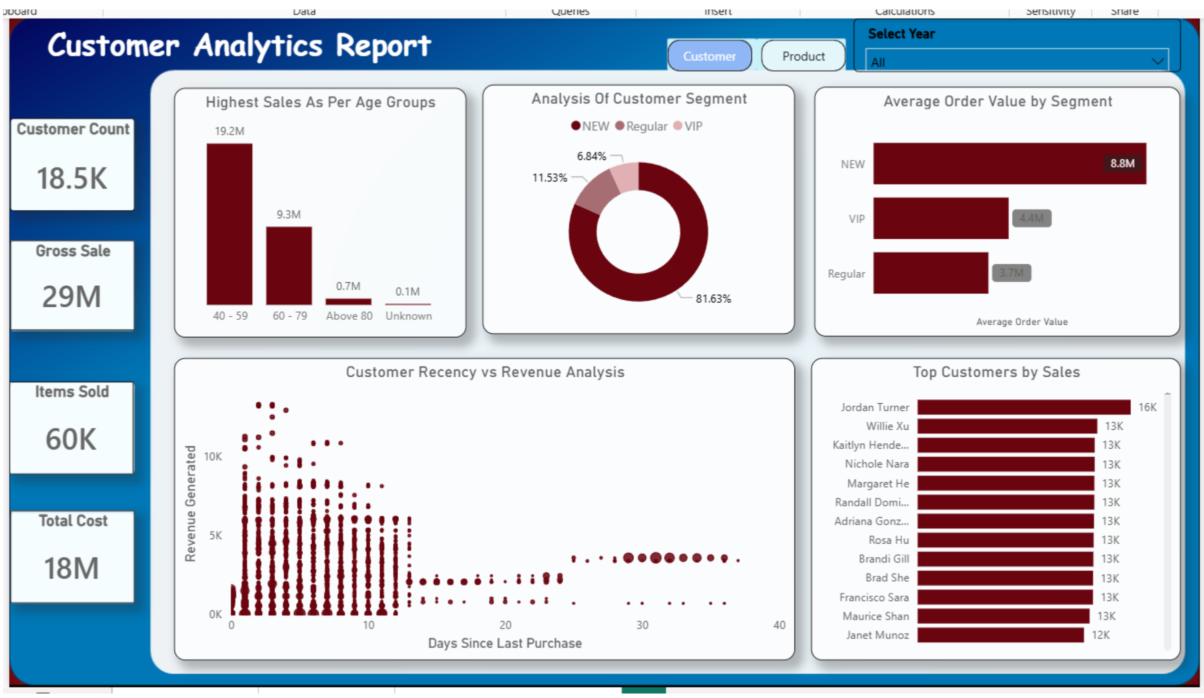
Strategic Insight:

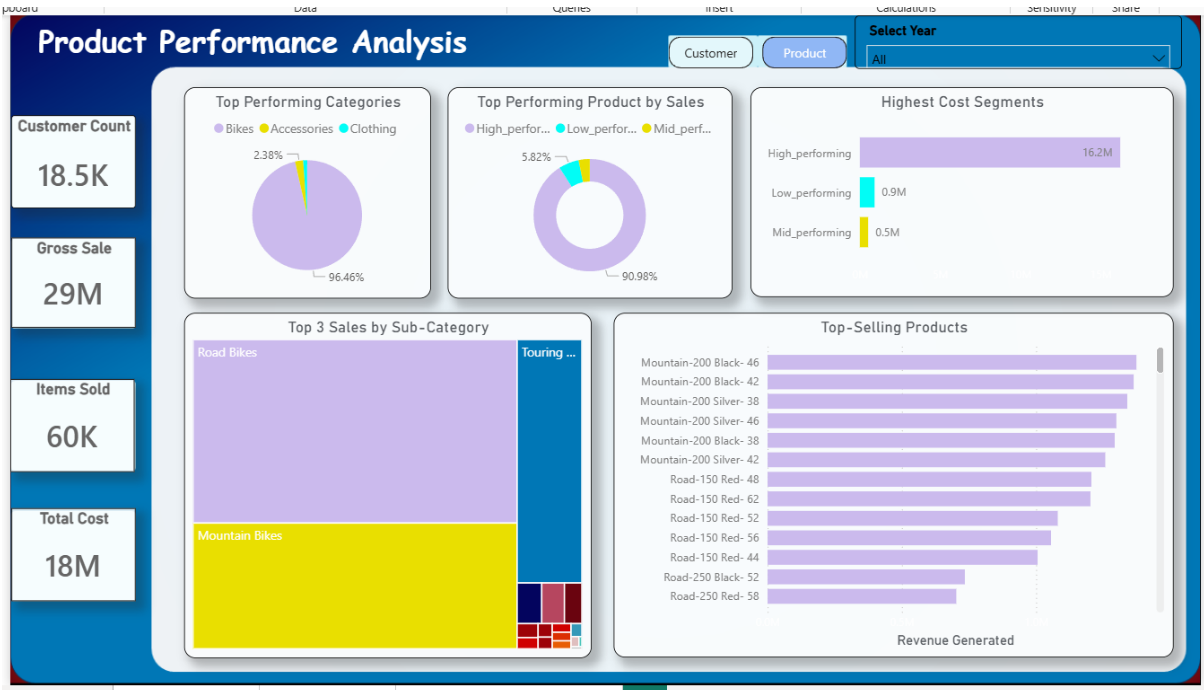
The company appears to have entered a more mass-market and utility-focused product strategy instead of premium specialization.

10.5 Overall 4-Year Business Analysis

❖ Business Evolution Summary

Year	Business Stage	Key Observation
2010	Premium Foundation	Small VIP customer base driving high-value sales
2011	Expansion Phase	Rapid customer growth with strong retention
2012	Market Transition	Shift from VIP to Regular customer dominance
2013	Mass Acquisition	Massive customer growth but reduced revenue efficiency





❖ Major Trends Identified

1. Customer Base Transformation

The business evolved through three stages:

- VIP-driven model
- Regular-customer model
- New-customer acquisition model

This shows the company continuously adapted its market strategy over time.

2. Product Preference Shift

Product dominance changed significantly:

- 2010 → Road Bikes
- 2011 → Road vs Mountain competition
- 2012 → Mountain Bikes dominance
- 2013 → Touring Bikes dominance

This indicates evolving customer preferences and changing market demand.

3. Revenue Efficiency Decline

Although customer count increased massively:

- Revenue growth did not scale proportionally
- 2013 showed lower sales despite higher customer volume

11. Key Business Insights

- A small percentage of customers generated the majority of revenue.
- Sales increased consistently across the analyzed years.
- Technology-related products performed better than other categories.
- Certain regions showed lower profitability despite high sales.
- Seasonal demand impacted monthly sales performance.
- Repeat customers contributed significantly to business growth.
- Some products required pricing or discount optimization.

12 Business Recommendations

Based on the analysis, the following recommendations are suggested:

Customer Strategy

- Focus retention efforts on high-value customers.
- Introduce loyalty programs for repeat buyers.
- Re-engage inactive customers through targeted campaigns.

Product Strategy

- Increase inventory for high-performing products.
- Review pricing strategies for low-profit products.
- Reduce excessive discounts impacting margins.

Regional Strategy

- Improve marketing in underperforming regions.
- Analyze operational costs affecting regional profitability.

Reporting Strategy

- Continue using dashboards for real-time monitoring.
- Track KPIs monthly for faster decision-making.

13 Conclusion

This project successfully transformed raw transactional data into actionable business insights using SQL and Power BI.

The analysis identified customer purchasing patterns, sales trends, top-performing products, and regional business opportunities. The dashboard developed during this project enables stakeholders to monitor performance efficiently and make informed business decisions.

14 Tools & Technologies Used

Tool	Purpose
SQL	Data Cleaning & Analysis
Power BI	Dashboard & Visualization
Excel	Initial Data Review
DAX	KPI Calculations
Power Query	Data Transformation

15 Future Improvements

Future enhancements for this project may include:

- Predictive sales forecasting
- Customer churn prediction
- Advanced customer lifetime value analysis
- Machine learning integration
- Automated reporting pipelines

16 Appendix

```sql
-- Find total revenue is generated by each customer

SELECT
	c.Customer_key,
	c.first_name,
	c.last_name,
	Sum(s.sales_amount) AS Total_Revenue
FROM gold.dim_customers c
Left JOIN gold.fact_sales s
		ON c.customer_key = s.customer_key
GROUP BY
	c.Customer_key,
	c.first_name,
	c.last_name
ORDER BY Total_Revenue DESC
```

```sql
-- What is the distribution of items sold across countries?

SELECT
	c.country,
	Sum(s.quantity) AS Quantity_sold
FROM gold.dim_customers c
Left JOIN gold.fact_sales s
			ON c.customer_key = s.customer_key
GROUP BY
	c.country
ORDER BY
	Quantity_sold DESC
```

```sql
-- Which 5 products generate the highest revenue?

SELECT TOP 5
	p.product_name,
	SUM(s.sales_amount) AS Total_Revenue
FROM gold.dim_products p
Left JOIN gold.fact_sales s
		ON p.product_key = s.product_key
GROUP BY
   p.product_name
ORDER BY
	Total_Revenue DESC
```

```sql
-- What are the 5 Worst performing product in terms of sales?

SELECT TOP 5
	p.product_name,
	SUM(s.sales_amount) AS Total_Revenue
FROM  gold.fact_sales s
Left JOIN  gold.dim_products p
		ON p.product_key = s.product_key
GROUP BY
   p.product_name
ORDER BY
	Total_Revenue
```

```sql
-- Find the top 10 customers who have generated the highest revenue?

SELECT TOP 10
	s.customer_key,
	c.first_name,
	c.last_name,
	SUM(s.sales_amount) AS Total_Revenue
FROM gold.fact_sales s
LEFT JOIN gold.dim_customers c
		ON s.customer_key = c.customer_key
GROUP BY
	s.customer_key,
	c.first_name,
	c.last_name
ORDER BY
	Total_Revenue DESC
```

(Complete SQL scripts are available in the GitHub repository:

https://github.com/bhaveshgurrap/sql-powerbi-customer-product-sales-analysis.git)